

## 100 Days of Azure DevOps

## Day 73 Handout — Instrumentation + connection string — dependencies tell you which friend is slow

8 - Monitoring & Observability · Personal lab only · Educational content · Not a sales pitch

### Architecture A — SDK, connection string, workspace-based resource

| Piece                      | Job                                              | Secret handling                                          |
|----------------------------|--------------------------------------------------|----------------------------------------------------------|
| App Insights resource      | Workspace-based component (classic is legacy)    | Create in rg-day73; tie to law-lab                       |
| Connection string          | SDK send target (InstrumentationKey is old news) | App setting or Key Vault reference. Never commit         |
| SDK / auto-instrumentation | Requests, exceptions, dependencies, live metrics | Runtime. A portal resource with no SDK is an empty GoPro |
| Dependencies               | HTTP/SQL/Redis map of who you wait on            | A slow app with clean CPU is often a slow friend         |
| Live metrics               | Incident stream                                  | Not the only alert; wallpaper is Day 74's problem        |

### Architecture B — instrument, demo data, or honest empty

| Constraint         | Default                                                                            | Honesty rule                                              |
|--------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Sample app you own | Add SDK or Azure auto-instrumentation; set connection string via app settings / KV | Do not paste the connection string into Git 'temporarily' |
| No app today       | Create the resource; walk Failures / Performance with demo or zero data            | Label demo data as demo. Empty is not 'we are reliable'   |
| Key Vault          | Store connection string as a secret; app uses @Microsoft.KeyVault(SecretUri=...)   | Pipeline need not print it (Day 65 door 3)                |
| Traffic            | Hit the app; fail on purpose once if you can                                       | Admiring an empty Failures blade                          |

### Architecture C — pitfalls that record nothing useful

| Pitfall                     | What you see                            | Fix                                                        |
|-----------------------------|-----------------------------------------|------------------------------------------------------------|
| Connection string in source | Git history is a camera with a postcard | Rotate; move to app settings / KV                          |
| Resource without SDK        | Charts exist, all zero, assumed healthy | Instrument or say 'dark'                                   |
| Sampling you did not know   | Missing a specific request              | Check sampling percentage before claiming 'never happened' |
| Live metrics as the pager   | Someone must stare                      | Day 74 alerts. Live metrics are for the incident           |
| Work app connection         | Employer telemetry in a personal lab    | Personal resource only                                     |

## One-liner to remember

**App Insights is a GoPro on your app — embarrassing, invaluable. No camera, no coaching. The connection string is a secret, not a README trophy.**

## Step-by-step lab (personal Azure / Azure DevOps only)

Personal rg-day73. Workspace-based Application Insights. Connection string never committed.

- 1 az monitor app-insights component create (or Portal) in rg-day73, linked to law-lab. Copy the connection string into the vault or a local password manager — not the repo.

- 2 Connect the sample app (SDK, Application Insights agent, or App Service auto-instrumentation). If you cannot, still create the resource and walk the blades with data honestly labeled.
- 3 Generate traffic. Open Failures and Performance. Click a dependency if one exists. Note one request duration.
- 4 Confirm git grep does not find InstrumentationKey or APPLICATIONINSIGHTS\_CONNECTION\_STRING values.
- 5 Optional: add the connection string as a Key Vault reference on an App Service you already have. App identity needs Secrets User.
- 6 Write docs/appi-day73.md: resource name, whether SDK or demo, that empty Failures means dark not reliable.

## Starter CLI + app setting — connection string stays out of Git

```
az monitor app-insights component create \  
  -g rg-day73 -l eastus -a appi-lab \  
  --workspace $(az monitor log-analytics workspace show -g rg-day71 -n law-lab --query id -o tsv)  
# Store the connection string in Key Vault, then:  
# az webapp config appsettings set -g rg-day73 -n <app> --settings \  
#   APPLICATIONINSIGHTS_CONNECTION_STRING=@Microsoft.KeyVault(SecretUri=https://<kv>.vault.azure.net/secrets/AppInsightsCs/)  
# KQL (once traffic exists):  
# requests | where timestamp > ago(1h) | summarize count(), avg(duration) by resultCode
```

## Done checklist

- App Insights resource exists and is workspace-based
- Connection string not in the repo
- Generated traffic or honestly labeled empty/demo
- Can explain dependencies vs live metrics vs failures map

## Tomorrow — Day 74

Alerts and action groups — screams need an inbox